

SECURITY REVIEW

FOR ARCHE



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01 - OVERVIEW

AUDIT SUMMARY

Arche engaged Kann Audits to perform a security review of the Arche smart contracts. From 28 – 30 April, a team of 3 security researchers performed a manual review of the in-scope contracts. All findings identified during the assessment are documented in this report.

TOTAL FINDINGS

0

ISSUES

CRITICAL	-
HIGH	-
MEDIUM	-
LOW	-
INFORMATIONAL	-

ENGAGEMENT DETAILS

PREPARED FOR	Arche
PREPARED BY	Kann Audits
DATE REPORTED	3 May 2026
REVIEW PERIOD	28 – 30 April

PROTOCOL OVERVIEW

Arche is a yield infrastructure protocol built on Ethereum. At its core is arUSD, a yield-bearing stablecoin fully backed by USDC. When you deposit USDC into the Arche vault, you receive arUSD, a composable ERC-4626 token that appreciates in value as the underlying strategies generate returns.

The protocol is designed around a single principle: maximize risk-adjusted yield on stablecoins without sacrificing liquidity or transparency. There are no lockups, no withdrawal delays, and no hidden mechanics. Every contract is on-chain and verifiable.

03 - WHAT WE REVIEWED

SCOPE

REPOSITORY	https://github.com/yieldarche/arche-contracts-audit
AUDITED COMMIT	ac1519f
FINAL COMMIT	ac1519f

#	FILE
01	src/Yearn V3 Vault.vy
02	src/HealthCheckAccountant.sol

04 - THE TEAM ON THIS ENGAGEMENT

AUDITORS

KA

Kann

SECURITY RESEARCHER

LY

Lyubo

SECURITY RESEARCHER

SA

Sang

SECURITY RESEARCHER

05 - HOW WE WORK

METHODOLOGY

An engagement with Kann Audits involves a team of security researchers performing independent reviews of the codebase at the start of the engagement, followed by collaborative review sessions where auditors share findings and explore additional attack vectors to improve overall coverage.

The review included a line-by-line manual inspection of the in-scope codebase, along with analysis of the protocol architecture, access control mechanisms and state transitions. The team also assessed the code against common smart contract security best practices and industry standards.

STEP BY STEP

OUR SECURITY PROCESS

01**Kickoff & Context**

A kickoff call to understand architecture, trust boundaries, threat assumptions and intended behaviour.

02**Scope & Preparation**

Confirm the commit, files in scope, dependencies and the riskiest areas. We open a private channel between your team and our auditors.

03**AI First Scan**

Kann AI, our proprietary model built to find security bugs, scans the codebase before the manual review starts.

04**Expert Security Review**

Line-by-line analysis of exploit paths, invariant breaks, edge cases, accounting and integration risks.

05**Findings & Proofs**

Each issue is confirmed, rated and backed by a proof of concept or a precise reproduction path.

06**Mitigation Review**

We review every fix, confirm it resolves the root cause and flag partial or incorrect mitigations.

COVERAGE

THE REVIEW SPECIFICALLY CONSIDERED

- ✓ Testing against common and uncommon attack vectors
- ✓ Contract logic matching the client's intended behaviour
- ✓ Manual line-by-line review by experienced auditors
- ✓ Architecture and design analysis for secure component interactions and clear trust boundaries
- ✓ Compliance with industry best practices and known smart contract security standards
- ✓ Cross-referencing patterns with industry-standard implementations
- ✓ Access control verification
- ✓ State transitions and logic, to detect inconsistencies, unexpected behaviour or exploitation paths

RISK CLASSIFICATION

Severity combines the **impact** of an issue if exploited with the **likelihood** that the conditions for exploitation are met.

● CRITICAL	Directly exploitable with few or no preconditions, leading to theft or permanent loss of funds or loss of control of the system. Must be fixed before deployment.
● HIGH	Directly exploitable vulnerabilities that can lead to significant material loss or critical impact on the system and require immediate fixing.
● MEDIUM	Moderate loss of funds or impact on multiple users, but only under specific conditions. Exploitable without privileged access, but typically requires certain assumptions or edge cases.
● LOW	Issues with limited impact that do not put funds directly at risk, such as edge cases or deviations from documented behaviour.
● INFORMATIONAL	Non-exploitable findings such as best-practice deviations, cosmetic issues or documentation and compliance notes. They do not need urgent remediation but improve code quality and maintainability.

07 - ALL ISSUES

FINDINGS SUMMARY

CRITICAL —	HIGH —	MEDIUM —	LOW —	INFO —
TOTAL 00 FIXED 00 ACKNOWLEDGED 00				

ID	TITLE	SEVERITY	STATUS
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DETAILED FINDINGS

During the audit of the Arche protocol, no critical, high, medium, or low severity vulnerabilities were identified within the scoped contracts.

DISCLAIMER

A security audit can never guarantee the complete absence of vulnerabilities. Audits are a time, resource and expertise-bound effort in which we aim to identify as many issues as possible.

This review covers only the files listed in the Scope section at the audited commit. Code changed after that commit, external dependencies, deployment configuration and operational practices may introduce additional risk.

ABOUT KANN AUDITS

Kann Audits is a Web3 security audit company helping DeFi protocols, blockchain networks, and infrastructure teams build securely at every stage, from development to launch and beyond.



EXPERT SECURITY AUDITS

Human-led smart contract security audits built around the architecture, business logic, integrations and risk profile of complex Web3 protocols.



AI SECURITY SCAN

Kann AI extracts Solidity call graphs and returns function-level security signals for expert validation.



FORMAL VERIFICATION

Define security properties for critical protocol behaviour and evaluate whether the implementation satisfies those specifications.



INCIDENT RESPONSE

A direct triage path for suspected exploits, critical vulnerabilities, suspicious transactions and urgent remediation review.



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WEB3 SECURITY AUDITS

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